The Positive-Chamber Koide Identity for Icosahedral Face Circulants

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Abstract

The stabilizer of an oriented icosahedral face acts regularly on its three corners, and every Hermitian response commuting with it is a circulant $C = aI + bR + \bar{b}R^2$. Where its eigenvalues are nonnegative and read as square-root masses, the Koide invariant is

$$Q = \frac{1}{3} + \frac{2}{3}\left(\frac{|b|}{a}\right)^2,$$

so $Q = 2/3$ is the balance $|b|/a = 1/\sqrt{2}$, and the phase of b sets the two mass ratios. A generation register in the coherent seam repair law supplies both: its branch weights $1/6 : 1/12 : 1/12$ give the balance, and its read count $2/9$ is the phase, kept over two million executed rounds. With no measured mass as input this gives $m_\mu/m_e = 206.770316$ and $m_\tau/m_e = 3477.4728$, at +9.83 ppm and +0.61 standard uncertainties from measurement. The register is a declared construction on the coherent law, so no source-only mass prediction is claimed. Under the balance and mass-ordering premises the measured electron and muon masses fix the tau mass inside a 72-eV enclosure centered at 1776.969027 MeV.

Keywords: Koide relation, circulant matrices, cyclic symmetry, icosahedral symmetry, finite event algebra, GNS representation

1 The face-corner carrier

Koide's charged-lepton relation [1] is usually written in terms of the three positive square-root masses. The construction below identifies the exact condition it imposes on a cyclic Hermitian response.

Let $G = A_5$, the alternating group on five letters, act by proper rotations on the icosahedron. The twenty outward oriented faces form the transitive orbit G/C_3. The stabilizer of one face cyclically

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permutes its three corners, so every face carries a local copy of the regular C_3 representation. If R denotes the cyclic shift, then

$$R^3 = I, \qquad R^\dagger = R^2.$$

The commutant of this regular action is the three-dimensional circulant algebra

$$\{x_0 I + x_1 R + x_2 R^2 : x_j \in \mathbb{C}\}.$$

Hermiticity restricts a response to

$$C = aI + bR + \bar{b}R^2, \qquad a \in \mathbb{R}, \quad b \in \mathbb{C}. \tag{1}$$

Write $b = \rho e^{i\delta}$, with $\rho = |b| \geq 0$. Fourier diagonalization gives

$$\lambda_k = a + 2\rho \cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2. \tag{2}$$

The unordered spectrum is independent of the chosen face representative. This is a local bundle statement. The sixty face-corner flags form the regular A_5 torsor; the geometry alone supplies no canonical global three-dimensional physical family space.

2 The positive-chamber identity

Theorem 1 (Positive-chamber Koide identity)**.** *Let C be the Hermitian circulant in Eq. (1), with $a > 0$. Suppose its three eigenvalues in Eq. (2) are nonnegative and, for one $s > 0$,*

$$\sqrt{m_k} = \sqrt{s}\lambda_k.$$

Then

$$Q := \frac{\sum_{k=0}^{2} m_k}{\left(\sum_{k=0}^{2} \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{\rho}{a}\right)^2. \tag{3}$$

Consequently,

$$Q = \frac{2}{3} \quad \Longleftrightarrow \quad \frac{\rho}{a} = \frac{1}{\sqrt{2}}. \tag{4}$$

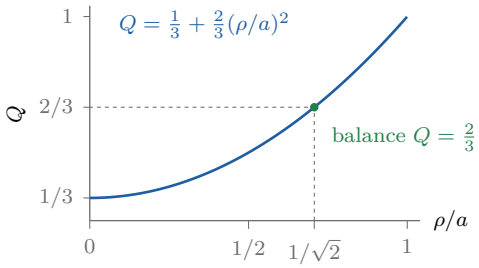
Proof. Set $c_k = \cos(\delta + 2\pi k/3)$. The roots-of-unity identities give

$$\sum_{k=0}^{2} c_k = 0, \qquad \sum_{k=0}^{2} c_k^2 = \frac{3}{2}.$$

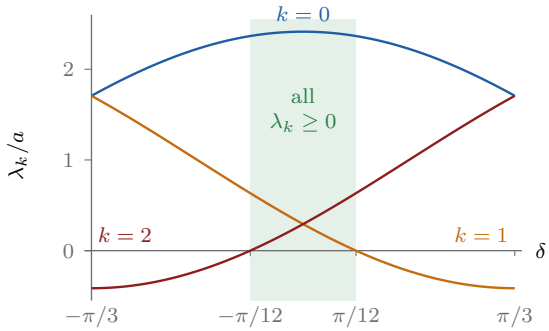
It follows that

$$\sum_k \lambda_k = 3a, \qquad \sum_k \lambda_k^2 = 3a^2 + 6\rho^2.$$

The scale s cancels from Q, yielding Eq. (3). Since $\rho/a \geq 0$, Eq. (3) equals $2/3$ exactly at $\rho/a = 1/\sqrt{2}$. □



(a) Koide ratio when $\sqrt{m_k} \propto \lambda_k \geq 0$



(b) $\lambda_k/a = 1 + \sqrt{2}\cos(\delta + 2\pi k/3)$ at balance; the positive chamber is $|\delta| \leq \pi/12$ modulo $2\pi/3$

Figure 1: (a) The Koide ratio of a nonnegative spectrum of $C = aI + bR + \bar{b}R^2$, $\rho = |b|$, as a function of ρ/a (Theorem 1); balance $\rho/a = 1/\sqrt{2}$ gives $Q = 2/3$. (b) At balance the three eigenvalues are nonnegative exactly in the chamber $|\delta| \leq \pi/12$ modulo $2\pi/3$, the chamber outside which the theorem makes no statement.

Remark 2 (The chamber is part of the theorem). The signed trace calculation holds algebraically for every δ. Physical square roots use $|\lambda_k|$ when an eigenvalue is negative, and then the denominator is not the signed trace. At $\rho/a = 1/\sqrt{2}$, the positive chamber is

$$|\delta| \leq \frac{\pi}{12} \pmod{\frac{2\pi}{3}}.$$

Theorem 1 makes no physical Koide statement outside that chamber.

Remark 3 (What the phase contains). Inside the positive chamber, Q is independent of δ. The three eigenvalues, and hence the two reported mass ratios, vary with δ. The balance condition removes one scale-free degree of freedom from a three-mass spectrum. It does not determine the ratios.

3 Conditional finite tracial balance

The balance condition in Eq. (4) can arise exactly from a finite event packet. This section states the packet as an independent conditional theorem. The packet realizes the target-informed balance premise through an asymmetric block choice, which is where that premise enters.

Let

$$\mathcal{V} = \mathbf{1} \oplus \chi \oplus \overline{\chi}, \qquad \mathcal{H}_{\mathrm{or}} = \mathbb{C}^2, \qquad \mathcal{A} = B(\mathcal{V} \otimes \mathcal{H}_{\mathrm{or}}) \simeq M_6(\mathbb{C}).$$

Here $\mathbf{1}$ is the neutral cyclic mode, $\chi \oplus \overline{\chi}$ is the two-dimensional charged plane, and $\mathcal{H}_{\mathrm{or}}$ is a two-state orientation record. Let P_0 and P_c project onto the neutral line and charged plane. For a rank-one oriented event q_+, define

$$Z_0 = P_0 \otimes I_{\mathrm{or}}, \qquad Z_c = P_c \otimes q_+, \qquad E_+ = Z_0 + Z_c.$$

Both Z_0 and Z_c have rank two.

Theorem 4 (Conditional finite tracial-GNS balance)**.** *Condition the normalized trace state $I_6/6$ on E_+. Map the orthonormal cyclic basis of the resulting square-root amplitude to (I, R, R^2) in $L^2(M_3(\mathbb{C}), \tau_3)$. Then*

$$p_0 = p_c = \frac{1}{2}, \qquad a = \sqrt{p_0}, \qquad \rho = \sqrt{\frac{p_c}{2}},$$

and therefore

$$\frac{\rho}{a} = \frac{1}{\sqrt{2}}, \qquad Q = \frac{2}{3}$$

whenever the positive square-root-mass reading of Theorem 1 is supplied.

Proof. Normalized-trace conditioning assigns probability proportional to block rank. Both accepted blocks have rank two, so $p_0 = p_c = 1/2$. In $L^2(M_3(\mathbb{C}), \tau_3)$, the operators I, R, R^2 are orthonormal. The canonical square-root vector therefore has neutral coefficient $\sqrt{p_0}$ and two charged coefficients of modulus $\sqrt{p_c/2}$. Their ratio is $1/\sqrt{2}$. Theorem 1 then gives $Q = 2/3$ on the positive chamber. □

Theorem 4 is finite and exact. Its event algebra, block choice, conditioned trace, and cyclic response map are premises. The block choice is asymmetric: the neutral line is paired with the full two-state orientation record and the charged plane with one oriented event. Treating the two blocks alike, both with I_{or} or both with q_+, gives $p_c/p_0 = 2$, hence $\rho/a = 1$ and $Q = 1$. The normalized trace fixes the probabilities only after the asymmetric event has been chosen, so the balanced modulus is the content of that choice. A physical charged-lepton statement additionally requires a quotient-visible map from the local face bundle to one chiral three-family response, preservation of the relevant block powers, and a mass readout. Those objects are absent from the theorem. The executable observer construction supplies none of them: the tau window consumes no native observer run, and its identification of balanced square-root masses with charged-lepton pole masses is a premise [8, 6].

Remark 5 (What a physical attachment would add)*.* The connected register fixes the normalization left free by a direct-sum maximum-entropy model; minimality of this response-local register is a declared hypothesis. A physical charged-lepton conclusion requires a chiral regular-C_3 carrier and faithful trace-preserving unital completely positive maps Φ, Ψ between the source and physical response algebras, with $\Psi\Phi = \mathrm{id}$. If they intertwine the block records and identify the physical GNS vector with $C = (Y_e^\dagger Y_e)^{1/4}$, Kadison–Schwarz makes Φ a $*$-monomorphism and an exact L^2 isometry, so it carries the equal source block powers to the physical response. For an accepted finite chiral checkpoint with three left and three right family modes, positive kinetic metrics, and a committed neutral Higgs direction of electroweak scale v, $\Phi = \mathrm{Ad}_{J_L \oplus J_E}$. With $M_F = X_F^2$, this gives

$$\widehat{Y}_e = \frac{\sqrt{2}}{v} J_L M_F J_E^\dagger, \qquad \mathcal{M}_L = J_L M_F J_L^\dagger.$$

An extended branch that adds graded physical completion and quotient source-law selection selects one dressed mass readout, given an exhaustive declared carrier class with a positive winner gap, a source-closed BV/BRST continuum, and an interval or contraction certificate for the charged QFT self-map. A balanced C_3 fixed point then gives $Q = 2/3$; a C_3-symmetric fixed point with charged attenuation $\chi_\star$ gives

$$Q = \frac{1 + e^{-2\chi_\star}}{3}.$$

An off-plane response requires the full response operator. None of these maps, checkpoints or certificates is constructed.

4 The balance and the phase from a generation register

The coherent seam repair event is the swap of the seam's port qubit (p, m), $p, m = (|a\rangle \pm |b\rangle)/\sqrt{2}$, with its record qubit, so the record holds the source endpoint relative to the address in its conjugate basis. The oriented record V changes the sign of record one at the reversed address and keeps the repair branch, which the plain record cancels.

Two seam objects share one face f: C, whose orientation is the screen orientation, and E, on a second edge of the hexagon around the f axis. Each has a load, a record and an address, and is prepared with activity $1/3$ and its load at the screen tail. A neutral register on the ports of f carries the face third-turn $R = \rho_f$. One tick is

$$U(\theta) = K_\theta\, T\, V\, (W_C \otimes W_E), \qquad K_\theta = e^{i\theta J},$$

where T applies $R^{\pm 1}$ when C repairs in orientation $\pm$ and J is the oriented current of C. The analyzer Φ is the Fourier zero of the three flags "E agrees, C idle", "C repairs $+$, E idle" and "C repairs $-$, E idle".

Theorem 6 (Register compression)**.** *The* 432*-state tick is unitary, and*

$$\langle \Phi | U(\theta) | p \rangle = \frac{\sqrt{2}}{3}\Big(\tfrac{1}{\sqrt{6}} I + \tfrac{1}{\sqrt{12}} \big(e^{i\theta} R + e^{-i\theta} R^{-1} \big) \Big).$$

Hence $|b|/a = 1/\sqrt{2}$ *and, in the positive chamber,* $Q = 2/3$ *for every* θ*.*

The squared coefficients $a^2 : |b|^2 : |b|^2 = 1/6 : 1/12 : 1/12$ are the probabilities of the agreeing hold of E and of the two oriented repairs of C. The register's exclusive read $[R_+] \oplus [H]$ has probability $1/12 + 1/6 - 2/72 = 2/9$. On every blank-record input with uniform address the seam program satisfies $\langle I - \mathrm{Re}\, U \rangle = P(\text{record } 1)$, so the energy clock reads read probabilities; N weak rounds within a tick give the zero-return operator $e^{-i\,(2/9)J}$ with error at most $1/N$, so the oriented current carries the phase $\theta = 2/9$ at unit clock time per tick. The declared clock fixes that unit: its edge term $(\pi/4)(I - S)$ reproduces the tick's root-swap gate exactly at clock times $1 + 4\mathbb{Z}$, with principal value one. A dressing that commutes with J leaves this phase unchanged at every order. The family mode k then has rate λ_k^2, and

$$\frac{m_\mu}{m_e} = 206.770316, \qquad \frac{m_\tau}{m_e} = 3477.4728.$$

The muon value lies 9.83 ppm above the measured ratio, which corresponds to a phase shift of -1.75×10^{-7}; the tau value lies 0.61 standard uncertainties above the measured ratio. Two-loop vacuum polarization by the lighter families, with each mass read at its own rest frequency, has the order and sign of the muon residual; the closest readings leave $+0.6$ ppm, and the constant of that term is open.

The count survives many ticks. One weak round is the tick of two cells, the kick on the oriented current, the inverse tick, one return test per cell and a reset after a negative test. Every round reads the first-tick weights, and N rounds per unit clock time give the phase

$$\theta_N = N \arctan \frac{P \sin(1/N)}{1 - P + P\cos(1/N)} = \tfrac{2}{9} - \tfrac{35}{2187 N^2} + O(N^{-4}), \qquad P = \tfrac{2}{9}.$$

Executed on the sixty-address carrier for 10^4 to 2×10^6 rounds, the phase follows this form and tends to $2/9$ within 10^{-13}. The four outcomes of the return tests, with weights $4/81$, $11/324$, $125/1296$

and 55/1296, are all kept: the coherence between two values j, j' of the current is multiplied per round by $1 - P + Pe^{-i(j-j')/N}$, so the phase holds on the complete record. The forward tick alone reads 2/9, 0.486, 0.536 and 0.467 at ticks one to four. The three characters of the face third-turn are the eigenvectors of the executed mass round, with rest rates

$$y_k = \tfrac{1}{27}\big(1 + \sqrt{2}\cos(\tfrac{2}{9} + \tfrac{2\pi k}{3})\big)^2 = 0.2096935560,\ 0.0124683656,\ 0.0000603006$$

for tau, muon and electron, summing to 2/9. With the electron mass as the only measured input they give $m_\mu = 105.659414$ MeV and $m_\tau = 1776.985$ MeV, against 105.6583755(23) and 1776.93 ± 0.09 MeV [2].

The register and analyzer are a declared construction on the coherent law. Open inside it are the operation that brings the pair onto one port, the source event of the kick, the rule that one partner is read per unit clock time, and the number N_p of phase rounds, which shifts m_μ/m_e by $-824/N_p$ at leading order [7, 8].

5 The conditional tau-mass window

Fixing $Q = 2/3$ in Eq. (3) and inserting the measured electron and muon masses turns the Koide coordinate into one quadratic in $\sqrt{m_\tau}$. The mass-ordering premise $m_\tau > m_\mu$ selects the physical root; the excluded second root sits at 3.317 MeV, below the muon mass. Propagating the one-sigma corners of the measured (m_e, m_μ) inputs [2] with outward rounding encloses the selected root in

$$m_\tau \in [1776.968991,\ 1776.969063]\ \text{MeV}, \tag{5}$$

a 72-eV window with center 1776.969027 MeV. The center sits 0.43 standard uncertainties from the measured 1776.93 ± 0.09 MeV, and the window is three orders of magnitude narrower than that measurement uncertainty. The dedicated Belle II 2023 measurement, $1777.09 \pm 0.08_{\text{stat}} \pm 0.11_{\text{syst}}$ MeV, lies 0.12097 MeV above the center, 0.89 of its combined 0.136 MeV uncertainty [4]. That comparison is retrospective, and its systematic uncertainty alone exceeds the 0.045 MeV compatibility floor, so it carries no verdict under the registered rule [5].

The comparison is governed by a registered conditional test whose target definition and decision rule were fixed in a frozen prediction record [3]: a world-average or dedicated-measurement central value more than three standard uncertainties from 1776.969027 MeV refutes the balanced-circulant premise; compatibility requires agreement within two standard uncertainties at a measurement uncertainty of at most 0.045 MeV; every other outcome carries no verdict. The balance premise, the equal-weight condition on the two response sectors of one icosahedral face circulant, is target-informed, so the window is a conditional postdiction whose evidential weight lies in the kill direction [3, 6]. Improving tau-mass measurements test the balanced-circulant premise and Koide's relation together. Both send every allowed (m_e, m_μ) to the same m_τ, so under one measurement model their likelihood ratio is one and no precision in this observable distinguishes them; the formula contains no screen-grain coordinate P and constrains no other OPH parameter [6].

6 Numerical diagnostic and provenance

Using the Particle Data Group 2026 central masses [2],

$$(m_e, m_\mu, m_\tau) = (0.51099895069,\ 105.6583755,\ 1776.93)\ \text{MeV},$$

gives

$$Q_{\mathrm{PDG}} = 0.6666644634026367, \qquad \rho/a = 0.7071044442682813, \qquad \delta = 0.2222247618945926.$$

The declared minimal complete public response (MCPR) law attenuates the balance and shifts the phase through three coefficients of a declared unified coupling u and pixel coordinate P:

$$\kappa = \frac{u/50}{1 + u/31 + u^2/310}, \qquad \chi = \frac{u^2/512}{1 - u/77}, \qquad \zeta = \frac{(P - \varphi)^2/(21\pi)}{1 - u/27 - u^2/135},$$

with $\varphi = (1 + \sqrt{5})/2$,

$$\frac{\rho}{a} = \frac{e^{-\chi}}{\sqrt{2}}, \qquad \delta = \frac{2}{9} + \zeta,$$

and a common multiplier chosen so that the product of the three masses divided by the cube of a declared electroweak scale v equals $e^{3\kappa}/(2 \cdot 6^{14})$. Inside the positive chamber, Eq. (3) gives the exact circulant identity

$$Q = \frac{1 + e^{-2\chi}}{3} \tag{6}$$

at every phase. The comparison packet of the particle construction [7] has

$$\begin{aligned} \chi &= 3.304899678958019 \times 10^{-6}, \qquad \zeta = 2.540957613468169 \times 10^{-6}, \\ Q_{\mathrm{MCPR}} &= 0.6666644634074956, \\ \rho/a &= 0.7071044442734351, \\ \frac{\rho/a - 1/\sqrt{2}}{1/\sqrt{2}} &= -3.3049 \times 10^{-6}. \end{aligned}$$

Its constants $50, 31, 310, 512, 77, 21, 27, 135$, amplitude, phase and determinant exponent are target-informed inputs, so the response is neither blind nor source-derived. The proximity of the two displayed Q values carries no significance or predictive weight.

Reading the three outputs, in the external energy unit E_*, as charged-lepton pole masses is an adopted physical premise, which similarity of dimensionless circulant coordinates does not derive. Under it the discrepancies are about -70.5 parts per million, almost entirely a common scale, and the muon/electron ratio differs by $+0.0725$ parts per million. The exact point $\rho/a = 1/\sqrt{2}$, $\delta = 2/9$ of the same circulant gives a muon/electron discrepancy of $+9.83$ parts per million. The particle paper gives the full comparison table, in which all twelve nonzero outputs lie outside their printed intervals, and no measured particle mass is derived [7].

Statement	Scope
Eq. (3)	exact algebra in the nonnegative-eigenvalue chamber
Theorem 4	exact implication conditional on the declared finite event packet
Icosahedral face fiber	exact local A_5/C_3 bundle and unordered spectrum
Tau window, Eq. (5)	exact quadratic consequence of the balance and ordering premises; registered decision rule
Physical family attachment	absent from the theorems; a premise of any physical reading
Phase and numerical ratios	balance leaves one phase controlling both scale-free ratios
Eq. (6)	exact circulant identity in the chamber
MCPR numerical proximity	target-informed comparison-only diagnostic
Charged pole comparison	adopted premise; outputs outside their printed intervals

7 Why residual symmetry does not finish the ratios

The family-shape problem can be stated in the real five-dimensional representation

$$W_5 \simeq \mathrm{Sym}^2_0(\mathbb{R}^3),$$

the traceless symmetric 3×3 matrices with action $A \mapsto gAg^T$.

Proposition 7 (Residual-stabilizer boundary)**.** *Invariance under a threefold or fivefold rotation about an axis n forces*

$$A = \alpha(nn^T - I/3),$$

which has a double eigenvalue. Invariance under a twofold rotation has a three-dimensional linear fixed locus, hence two projective parameters, and that locus admits simple spectrum.

Proof. Choose the rotation axis as the third coordinate. Decompose a traceless symmetric matrix into a planar scalar, planar spin-two anisotropy, transverse vector, and axial entry. Rotation by θ acts on the anisotropy by 2θ and on the transverse vector by θ. For $\theta = 2\pi/3$ or $2\pi/5$, neither block has a nonzero fixed vector, so only the axial combination remains. For $\theta = \pi$, the planar symmetric block survives:

$$A = \begin{pmatrix} u & v & 0 \\ v & w & 0 \\ 0 & 0 & -u-w \end{pmatrix}.$$

This space has dimension three and contains matrices with three distinct eigenvalues. □

Threefold and fivefold symmetry cannot produce three distinct masses. Twofold symmetry leaves two scale-free coordinates, exactly enough to carry the two mass ratios without selecting them. A numerical spectrum therefore requires a specific invariant potential derived from the screen dynamics. This proposition locates the missing input; it supplies no replacement fit.

8 Formal proof boundary

The algebraic core has a Lean 4 formalization. It proves the root sum, square sum, quotient formula, reciprocal-square-root identity, and the nonnegative-modulus equivalence

$$\frac{1}{3} + \frac{2}{3}r^2 = \frac{2}{3} \quad \Longleftrightarrow \quad r = \frac{1}{\sqrt{2}}.$$

The formal module deliberately does not identify signed eigenvalues with physical square roots. Positivity of all three eigenvalues, the phase selection, and the source-to-charged-family attachment are outside its statement. This division matches Theorem 1: the checked algebra is exact, and the physical reading is a separate premise.

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AI Assistance Disclosure

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